

Management Guidelines for Atopic Dermatitis in Children

Determining Atopic Dermatitis (AD) Severity

The extent and severity of AD can be determined by

1) Grading the extent of the affected areas

2) Defining the severity: induration, erythema, excoriation, lichenification, scaling, oozing, and crusting.

Four signs of eczema used to calculate severity score

Intensity	None	Mild	Moderate	Severe
Redness	 Score 0	 Score 1	 Score 2	 Score 3
Thickness/swelling	 Score 0	 Score 1	 Score 2	 Score 3
Scratching	 Score 0	 Score 1	 Score 2	 Score 3
Lichenification/prurigo	 Score 0	 Score 1	 Score 2	 Score 3

<http://www.dermnetnz.org/dermatitis/easi.html>

For daily clinical practice, this classification can be used instead

Mild: mainly dry skin, few thin patches only

Moderate: fold eczema (elbows, knees, wrists and ankles) + neck + face

Severe: extensive involvement and signs of increased lesion severity (erythema, papulation, excoriation)

Other clinical measures of severity include

- Significant itching affecting quality of life
- Poor sleep and daily functioning
- Persistence of disease despite treatment

Authors: Elizabeth Tham, David Ng, Bernice Oh and Alison Lee

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Management in the Children's Emergency/Outpatient Specialist Clinic

Mild Basic AD management

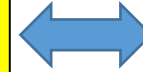
1. Moisturize at least twice daily (CREAMS, not lotion)
2. Soap-free bath wash and shampoos
3. Reduce skin bacterial load*
4. Consider low potency topical steroids (Class V-VII)
 - Hydrocortisone 1% (face)
 - Mometasone furoate 0.1% cream (Elomet) or Betamethasone valerate 0.1% (body)
5. Trigger avoidance (soaps, wool, temp extremes and proven allergens)



Moderate

Basic AD management PLUS

1. Increase moisturizing to at least 3-4x daily
2. Medium potency topical steroids (Class III-IV) on all inflamed lesions
 - Mometasone furoate 0.1% cream (Elomet)
 - Desonide 0.05% lotion (face)
3. Sedating antihistamines for itch
4. Refer to Allergy Team at next available date



Severe

Moderate AD management PLUS

1. Liberal application of moisturizers $\geq 5x/day$
2. Consider Oral Prednisolone 1mg/kg/day for 3-5 days
3. Mid to high potency steroids (Class I-III)
 - Elomet Ointment 0.1% or Betamethasone dipropionate 0.05% cream (body)
 - Elomet 0.1% cream (face)
4. Consider oral/IV antibiotics if infected (NOT topical antibiotics)
5. Refer to Allergy Team within 3-5 days if admission not required

Escalate treatment in an acute flare

Review after 1 week and step down if status has improved

*see below for antiseptic therapy options

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Indications for admission

- 1) Bacterial infection with fever or in an unwell child
- 2) Poor oral intake
- 3) Inability to tolerate oral medications (eg oral antibiotics)
- 4) Eczema Herpeticum
- 5) Superimposed varicella infection
- 6) Inability of caregiver to cope with skin care at home

Referral and review

- 1) All patients with infected eczema should be followed up within 2-3 days for early review
- 2) Early Referral to the Allergy team should also be considered in
 - Moderate to severe AD
 - If food allergy is a concern or if parents are eliminating foods without prior allergy assessment
 - Patients who have more than 20% skin involvement (or 10% skin involvement if affected areas include the eyelids, hands, or intertriginous areas) that has not been responsive to first-line treatment
 - Regular use of topical steroids more potent than Class V or VI.
 - Any prior admission for eczema

Red Flags

Bacterial Superinfections

Clinical features

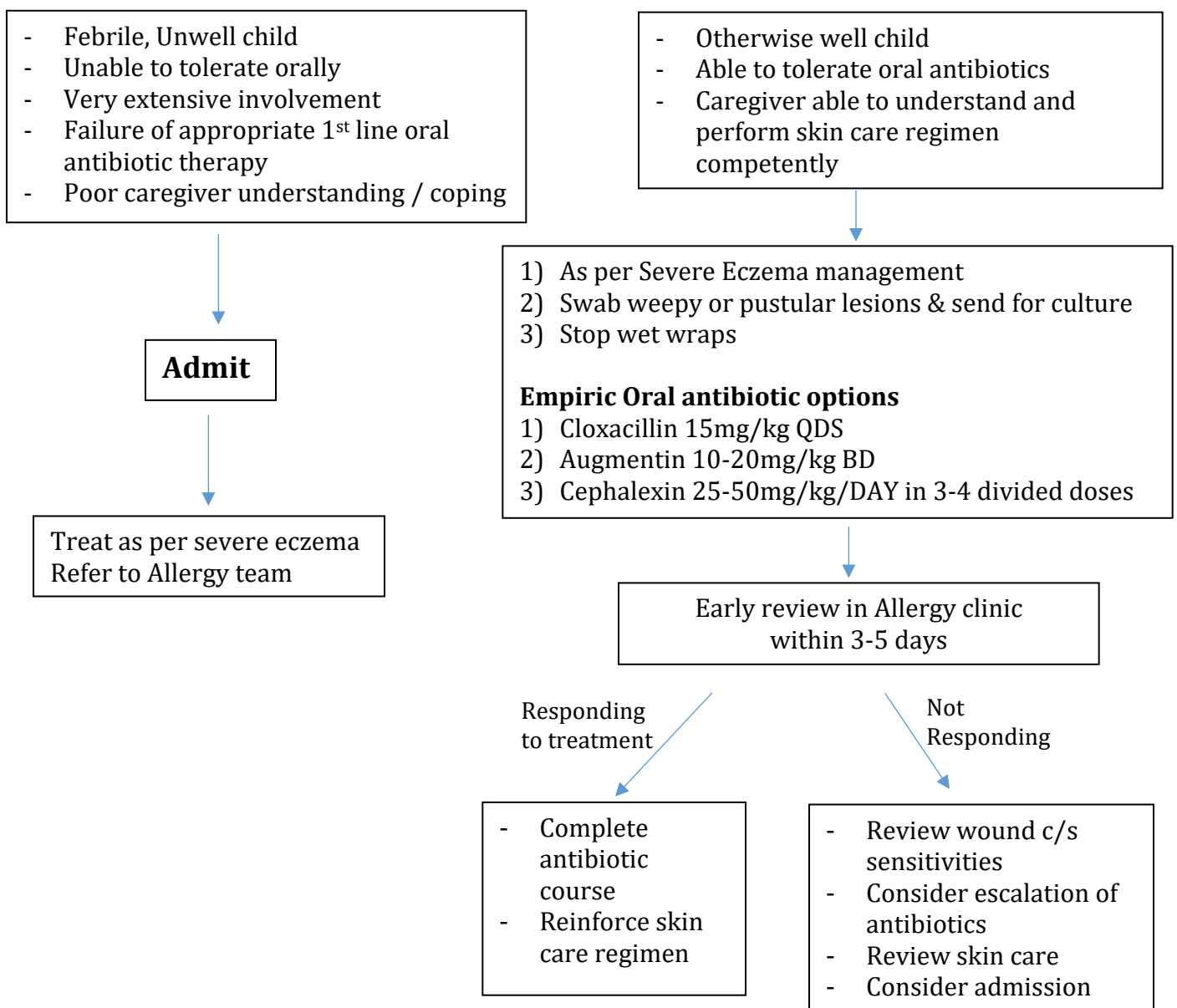
- Weepy lesions
- Purulent discharge, pustules
- Honey crusted lesions (impetigo)

Typical organisms – Staph Aureus, Streptococcus

Complications

- Cellulitis, abscesses
- Osteomyelitis, septic arthritis
- Sepsis, endocarditis

Management workflow



Eczema Herpeticum

Secondary infection of AD skin with herpes simplex virus (HSV)

Clinical features

- Typically painful, punched out erosions or ulcers
- Papulovesicles may be present in the early stages
- May also have fever, malaise & lymphadenopathy
- Secondary bacterial infection is also common
- **Complications : keratoconjunctivitis, disseminated infection and multi-organ involvement**



<http://www.pediatricsconsultantlive.com/pediatric-skin-diseases/eczema-herpeticum-0>

Admission for IV Acyclovir is recommended in all cases as rapid progression can occur and the bioavailability of oral acyclovir is low (15-30%)

Management

- 1) Take swab from freshly de-roofed vesicle or ulcer base
Use the Universal transport medium swab (shown below)
- 2) Send for HSV PCR
Aurora test " HSV (Herpes simplex virus) DNA"



- 3) Treatment
 - IV Acyclovir 5 mg / kg q8h (ensure good hydration)
Total treatment duration: 7-10 days
 - Pain control
 - Identify and treat secondary bacterial infection
 - Avoid topical steroids and calcineurin inhibitors to actively herpes-infected areas until lesions heal
 - Avoid systemic immunosuppressives during acute infection
 - Ophthalmologic consult if facial involvement close to eye area
 - Refer to Allergy Team for skin care advice

Specific Treatment Advice

1. Bath-time and skin care	- Avoid common irritants: sweat, soaps, shampoos, detergents, perfumed products - Cool to warm bath daily to twice per day using soap-free bathwash - Dab dry with towel (do not rub) after showering - Generously apply a moisturizer after bathing/cooling down - After exercise/sweating – rinse body with water and dry off - Wear loose, cotton clothing and socks which do not retain sweat (avoid tight fitting, synthetic materials, and wearing shoes without socks)
2. Moisturize	- Generous application, especially after baths - If night-time itch present consider one application before bedtime - Moisturizing CREAMS (non-perfumed) should be used (not oils, ointments or lotions) - Available in NUH: Atopiclair[®], Balneum Intensive Cream, Cetaphil[®], Cetaphil[®] RESTORADERM[®], Dermaveen[®], Physiogel[™], Physiogel[™] AI, QV cream, QV flare up, Sebamed[®],
3. Reducing bacterial load	- Chlorhexidine solution (0.05%): dab on affected areas, or put in bathtub and soak daily - Zinc-Copper-Sulphate Solution: Dilute 1ml to 10ml water and dab on affected area or use as last rinse BD (also acts as an anti-inflammatory agent) - Dilute bleach baths 2x/week as maintenance therapy in moderate-severe eczema - Swimming pools (must wear sun block & shower after swimming) - Potassium permanganate: apply to affected areas bd-tds (usually in-patient)
4. Allergen avoidance	<u>Food Allergens</u> - Only around 10% of children with severe eczema have a food allergen trigger - Allergy evaluation should be in consult with an allergy specialist and requires a positive skin prick test or specific serum IgE together with a suggestive clinical history of food-related exacerbations - Food elimination without prior allergy evaluation is NOT recommended - Pet avoidance (rare): only if confirmed on allergy testing

	<u>House Dust Mites</u> The use of HDM avoidance measures in prevention of AD flares is controversial as published trials show conflicting results. Expert consensus statements have thus proposed that skin prick testing and avoidance advice may be given only in these circumstances: 1) In sensitized patients (with AD) with concurrent asthma or perennial rhinitis triggered by dust mites 2) When there is a <u>clear clinical history</u> of HDM induced exacerbations of AD in sensitized patients 3) In patients who are sensitized to HDM and whose AD is otherwise uncontrolled Basic house dust mite avoidance advice: 1) No carpets or upholstery in bedroom 2) Keep clothes in closed cupboard with doors 3) Wash bed sheets in hot water above 60°C (once per week) 4) Soft toys can be placed in zip lock bag and freeze overnight 5) Vacuum mattress regularly 6) Encourage outdoor activities and stay out of the bedroom as much as possible There is current no evidence for the efficacy of sublingual immunotherapy (SLIT) in atopic dermatitis
5. Sedating antihistamines	- To improve quality of life and allow sound sleep at night. - Non-sedating antihistamines have little role in the control of itch in AD, as it is not histamine related.
6. Topical steroids	- For face and scalp: Class VI-VII - For body: can consider higher class: Class IV – V - High potency steroids should not be used for prolonged periods - Dosing: apply thinly BD x 1 week and review

Topical Steroids Classification

Class	Examples
I (Most potent)	Betamethasone dipropionate 0.05% ointment Clobetasol propionate 0.05% cream or ointment
II	Betamethasone dipropionate 0.05% cream Mometasone furoate 0.1% ointment
III	Betamethasone dipropionate 0.05% cream or ointment *Betamethasone valerate 0.1% ointment *Fluticasone propionate 0.005% ointment *Triamcinolone acetonide 0.1% ointment
IV	Fluocinolone acetonide 0.05% ointment Mometasone furoate 0.1% cream or lotion *Triamcinolone acetonide 0.1% cream
V	Betamethasone valerate 0.1% cream Fluocinolone acetonide 0.025% cream Fluticasone propionate 0.05% cream
VI	Betamethasone valerate 0.1% lotion (scalp) *Desonide 0.05% lotion *Fluocinolone acetonide 0.01% cream/lotion *Triamcinolone acetonide 0.025%
VII (Least potent)	Hydrocortisone 1% cream

*NOT found in NUH

Other options (with Allergy team consultation)

2. Topical calcineurin inhibitors (e.g. Protopic Ointment 0.03% or Elidel® 1%)

- a. Can be used in children more than 2 years old with moderate or severe AD as a steroid-sparing alternative
- b. Parents should be informed of the black-box warning, and possible side effects such as the initial burning/tingling sensations which may be experienced

3. Wet wraps

- a. Children with moderate to severely affected areas may require wet wraps both for relief of itch and suppression of acute flares
- b. Not advised if infected and on weepy areas
- c. Refer to Allergy nurse for education

4. Consider referral to allied health staff for **non-pharmacological management & support where relevant**

- a. Play therapist
- b. Child Psychologist
- c. Dietician
- d. Medical Social Worker

References:

- 1) Eichenfield LF, Tom WL, Berger TG, et al. Guidelines of care for the management of atopic dermatitis: section 2. Management and treatment of atopic dermatitis with topical therapies. *J Am Acad Dermatol*. 2014 Jul;71(1):116-32 (American Academy of Dermatology)
- 2) Sidbury R, Tom WL, Bergman JN, Cooper KD, et al. Guidelines of care for the management of atopic dermatitis: Section 4. Prevention of disease flares and use of adjunctive therapies and approaches. *J Am Acad Dermatol*. 2014 Dec;71(6):1218-33. (American Academy of Dermatology)
- 3) Arkwright PD, Motala C, Subramanian H, et al. Atopic Dermatitis Working Group of the Allergic Skin Diseases Committee of the AAAAI. Management of difficult-to-treat atopic dermatitis. *J Allergy Clin Immunol Pract*. 2013 Mar;1(2):142-51.
- 4) Schneider L, Tilles S, Lio P, Boguniewicz M, Beck L, LeBovidge J, and Novak N. Atopic dermatitis: A practice parameter update 2012 *JACI* 2013: 131;295-299.e27
- 5) Charman CR, Venn AJ, Williams H, Bigby M. Measuring Atopic Eczema Severity Visually: Which Variables Are Most Important to Patients? *Arch Dermatol*. 2005;141(9):1146-1151. doi:10.1001/archderm.141.9.1146.
- 6) Hanifin, J. M., Thurston, M., Omoto, M., Cherill, R., Tofte, S. J., Graeber, M. and Evaluator Group, T. E. (2001), The eczema area and severity index (EASI): assessment of reliability in atopic dermatitis. *Experimental Dermatology*, 10: 11–18. doi: 10.1034/j.1600-0625.2001.100102.x
- 7) NICE clinical guideline 57: Management of atopic eczema in children from birth up to the age of 12 years
- 8) DermNet New Zealand guidelines
<http://www.dermnetnz.org/dermatitis/easi.html>